

Object classification and visualization with edge artificial intelligence for a customized camera trap platform

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ABSTRACT

The camera traps have revolutionized the image and video capture in ecology and are often used to monitor and record animal presence. With miniaturization of low power electronic devices, better battery technologies, and software advancements, it has become possible to use the edge devices, such as Raspberry Pi as camera traps that can not only capture images and videos, but can also enable sophisticated image processing, and off-site communications. These developments can help to provide near real-time insights and reduce the manual processing of images. The on-board image classification and visualization is facilitated by the advancements in the Deep Neural Networks (DNN), transfer learning approaches, and software libraries.

This paper provides an investigation of image classification with transfer learning approaches using pre-trained DNN models, and visualizations with Explainable Artificial Intelligence (XAI) techniques on Raspberry Pi Zero (RPI-Z) edge device. The MobileNetV2 model was used for image classification on the Florida-Part1 dataset obtaining the results for precision, recall, and F1-score as 0.95, 0.96, and 0.95 respectively. We also compared the model performance of MobileNetV2, EfficientNetV2B0, and MobileViT models for classification on the Extinction dataset with the best results for precision, recall, and F1-score as 0.97, 0.96, and 0.96 respectively, obtained with the EfficientNetV2B0 model. Two XAI techniques, Gradient-weighted-Class Activation Mapping (Grad-CAM) and Occlusion Sensitivity were used for visualization through heatmaps, to highlight the relative importance of the image areas contributing to the DNN model's prediction, that can also help to understand the model's performance and bias. The results provide practical use case scenarios for utilizing the transfer learning approaches, model optimization and deployment to edge devices, and model visualizations in ecological research.

1. Introduction

Camera traps are now commonplace in ecological studies and have made it easier to acquire image and video data for biodiversity (Chalmers et al., 2023; Zhu et al., 2022), animal behaviour (Petso et al., 2021; Schindler and Steinhage, 2021), animal re-identification (Kuncheva et al., 2023) conservation (Dujon and Schofield, 2019), distribution (Ryo et al., 2021), and population estimation (Simoes et al., 2023). The ecological studies often cover large geographical areas, longer study timeframes, and thousands of camera traps that can generate millions of images. This faster rate of image data accumulation is difficult to manage and analyze manually (Chalmers et al., 2023; Duggan et al., 2021; Dujon and Schofield, 2019; Norouzzadeh et al., 2018). For the captured image data to be of timely use, requires a move from the

traditional manual classification methods to automated image processing techniques. The automated processing of the image can take place off-site by transmitting the images (Chalmers et al., 2023; Nazir et al., 2017a; Whytock et al., 2023), retrieving the camera trap data (Wildlife Insights, n.d), or can take place on-board the device (Nazir et al., 2017b; Wei et al., 2020) which can also be accompanied by transmitting, subset of images, or the class labels from the deployment site (Whytock et al., 2023; Zualkernan et al., 2022). The real-time classification of images acquired in the camera trap studies can aid many application use cases, such as to avoid human-wildlife conflict or poaching, facilitating timely and informed decisions for preserving the environment and ecosystem.

Animal classification studies are increasingly utilizing the DNN models which have shown remarkable performance on challenging datasets. Animal behaviour monitoring using videos from a wildlife

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crossing showed an average precision of 63.8% for animal identification (Schindler and Steinhage, 2021). The animal recognition and subsequent re-identification is important for the endangered ecosystems and climate change concerns (Kuncheva et al., 2023). A trained DNN model comprises of the model parameters or layer weights learnt during the training process. The training of large DNN models can take days or weeks on state-of-the-art computers, whereas the inference only takes few seconds even on an edge device. Many researchers would not have the time, labelled training data, requisite skills, and computing resources to train a model from scratch.

The technical challenges of processing camera trap data can be resolved using AI frameworks, specially designed for this purpose, with some platforms also allowing users to modify the workflows (Velez et al., 2023). Camera trap images for animal detection used the MegaDetector model, and it was found to perform exceptionally better compared to the manual process (Leorna and Brinkman, 2022). Wildlife identification in camera trap videos used pre-trained MegaDetector with Faster R-CNN and Inception-ResNet-v2 for image classification, and achieved 73.92% Mean Average Precision (mAP) (Simoes et al., 2023). Another technique that has gained prominence is transfer learning, in which a pre-trained model trained on a large dataset, can be used in other classification problems. The transfer learning approaches are increasingly being used due to the lack of skills required for developing a machine learning model from scratch (Dufourq et al., 2022). Passive acoustic monitoring (PAM) of Hainan gibbon (*Nomascus hainanus*) showed that transfer learning using 12 pre-trained models, obtained good performance, required fewer implementation details, and was easier to use (Dufourq et al., 2022). The access to the pre-trained models (TensorFlow Hub, n.d) and similar frameworks (Stancic et al., 2022) has made it easier to use transfer learning approaches. The pre-trained models can be used with no change, or some layers of the pre-trained model can be tuned or re-trained, a process termed as fine-tuning. Based on the application needs, some studies select a large size pre-trained model, such as ResNet50 (98 MB), or comparatively smaller model, such as MobileNetV2 (14 MB), useful for low power edge devices.

The transfer learning of large sized DNN models has been extensively used in ecological studies. A study for the species identification, selected a subset of 26 out of 48 species classes (based on their relative representation in the dataset) from the Snapshot Serengeti dataset, and achieved Top-1 accuracy of 35.4% on the unbalanced dataset, whereas for the balanced dataset the accuracy was 88.9% (Villa et al., 2017). Animal identification, count, and behaviour were inferred from Snapshot Serengeti dataset, using nine DNN models, achieving a classification accuracy of 93.8% (Norouzzadeh et al., 2018). Transfer learning on the entire dataset was found to have limited value, however, it was concluded to be useful for projects with smaller dataset (Norouzzadeh et al., 2018). The high classification performance reported in the study (Norouzzadeh et al., 2018) was attributed to selecting single species images, and manual cropping and segmenting of animal images (Westworth et al., 2022). The effect of data size, unbalanced data, and model generalizability for species recognition were investigated with transfer learning of six DNN models on the Parks Canada dataset comprising 47,279 images representing 55 animal species, and achieved the best accuracy of 95.6% accuracy with DenseNet201 (Schneider et al., 2020). The performance on small scale datasets was found good, however, the model accuracy decreased on images from different locations, and with more images in underrepresented image classes (Schneider et al., 2020). Species recognition with transfer learning was shown to produce better results compared to training from scratch, however the classification accuracy was low for animals with less numbers in the dataset (Willi et al., 2019). The species identification of ten wild mammal species used pre-trained FasterRCNN+InceptionResNetV2 model (TensorFlow Hub) without training, to classify ten images of each species, and achieved a classification accuracy of 71% (Carl et al., 2020). Transfer learning was used for a small dataset comprising 17 object classes with 275 images per class, achieving an accuracy of 92% (Duggan et al., 2021). The

authors concluded that transfer learning is useful where the species are easily distinguishable in the images (Duggan et al., 2021). The ResNet50 model pre-trained on the ImageNet dataset (ImageNet Data, n.d), was used for image classification on a dataset of 26 African mammals (Whytock et al., 2021). The entire image was used for training without any bounding boxes and it was shown that by selecting an appropriate model's classification threshold, better classification accuracies can be obtained, reducing the need for manual classification (Whytock et al., 2021).

Some studies have used transfer learning approaches with small sized models, such as MobileNetV2, which had comparable performance to the large models (Schneider et al., 2020). Binary classification of Eurasian Lynx (*Lynx lynx*) images from a camera trap study were used to compare the performance of 36 pre-trained DNN models, including MobileNetV2 and MobileNetV3, on an unbalanced dataset of 293,604 images, and found the less complex models to be less accurate (Stancic et al., 2022). The recent improvements to the DNN architectures designed for mobile devices, with reduced model sizes, memory, and processing requirements, have made it possible to run state-of-the-art DNN models on mobile devices, or Raspberry Pi (Raspberry Pi, n.d) like devices in the field.

The open access to the transfer learning techniques for image classification can substantially reduce the number of images to be manually processed. However, a DNN model is like a black box, that is with its internal working hidden, there is a lack of understanding of how and why it made a particular prediction, e.g., a class label for classification problems. XAI is an emerging research area that can help to generate heatmaps for the visualization of the model decision, aiding validation of the model's classification decision. This could be helpful in understanding bias, increasing generalizability, and performance (Dwivedi et al., 2023).

In this paper, we have used MobileNetV2 due to its small size, desirable in edge applications and its extensive use (Cunha et al., 2021; Petso et al., 2021; Yang et al., 2022), however, we have also provided comparison of MobileNetV2 model with other state-of-the-art models. The focus of this study is to explore the use of different transfer learning models and approaches for animal classification on-board a prototype camera trap. The contributions of this paper are as follows:

- Comparison of transfer learning with fine-tuned approaches for image classification on challenging image datasets
- Visualization of the image classification results using two Explainable AI techniques for better understanding of the model and its predictions
- Design of a solar-powered prototype camera trap platform using Raspberry Pi Zero (RPI-Z) for on-board image classification and visualization
- Comparison of four state-of-the-art small size DNN models, including vision transformers for image classification using transfer learning and exploring the role of hyperparameter optimization on model performance

Rest of this paper is structured as follows: The related work is provided in Section 2. Section 3 covers the materials and methods. The results and discussion are provided in Section 4 and 5. Finally, conclusion and future work are provided in Section 6.

2. Related work

We provide the related work aligned with the three focus areas. Firstly, we cover the studies aimed at using DNN models with transfer learning techniques using camera traps on edge devices. Secondly, the use of transfer learning with fine-tuning of small models is covered, and thirdly, we have included the studies that use XAI techniques for image visualization. A summary of the selected approaches is provided in Table 1.

Table 1
Summary of transfer learning approaches and XAI techniques.

Study	Method	Dataset	DNN model	Performance	Reference
Raspberry Pi 4, Nvidia Jetson, Google Coral Board for wildlife monitoring and real-time transmission	Transfer learning	66,400 images of 15 classes, Emirates Nature WWF	TensorFlow Lite InceptionV3, MobileNetV2, ResNet18, EfficientNetB1, DenseNet121, and Xception pre-trained on ImageNet neural network models	96.1% accuracy, F1-Score of 0.87 with Xception model	Zualkernan et al., 2022
Animal recognition of 22 species	Filtering camera trap images, precision and latency on edge devices	Caltech camera trap dataset, 243,100 images, Snapshot Serengeti 78,000 images	MobileNetV2, EfficientNetB0, and EfficientNetB3	87.67% precision	(Cunha et al., 2021)
Real time alerts from camera trap with a Raspberry Pi	Image labels transmitted over Iridium Satellite network	Species classification	Google AutoML	588 image alerts	(Whytock et al., 2023)
Bird classification of ten species	Images transmitted over cellular network	34,970 images	Faster-RCNN	Sensitivity 88.79%, Specificity 98.16% and accuracy 96.71%	(Chalmers et al., 2023)
Passive acoustic monitoring	Twelve pre-trained models	Multiple datasets	Fine tuning of models, including MobileNetV2	82% F1 score	(Dufourq et al., 2022)
Species distribution model	Relationship between the species occurrence data and the environmental data	African elephant (<i>Loxodonta Africana</i>) dataset	XAI technique for machine learning models	Understanding the model prediction	(Ryo et al., 2021)

2.1. Transfer learning with edge devices

The use of DNN models for processing camera trap images can take place on-board the camera trap, or the images could be transmitted to off-site servers or cloud platforms. Faster-RCNN model was used for an automatic real-time bird classification system. The images captured by the camera trap were communicated over the cellular network to be processed, hence overcoming the delays in manually collecting the images from the site (Chalmers et al., 2023).

Some studies have used edge computing for animal classification in the field conditions. The images captured from a commercial camera trap were classified using six pre-trained DNN models including MobileNetV2 and EfficientNetB1, running on a RPi 4, with the image labels transmitted over the Long-Range Wide Area Network (LoRaWAN) to the cloud platform for wider accessibility. RPi had a latency of 2.83 sec/image but consumed the least average current, 838.99 mA compared to the other edge devices (Zualkernan et al., 2022). The DNN models were trained using 66,400 images from Emirates Nature - World Wide Fund for Nature (WWF) dataset, achieving an accuracy of 96.1% with Xception model (Zualkernan et al., 2022). RPi 3 was connected to camera traps and a classification accuracy of 87.6% was achieved with ResNet-18 and ResNet-50 models (Mathur and Khattar, 2019). The subset dataset comprised of three classes and 64,000 images, selected from a dataset with 24 classes and 200,000 images (Mathur and Khattar, 2019). An open source camera trap was developed using RPi-Z for plant-insect interactions (Droissart et al., 2021). The platform was used to record continuous video for around 72 hours with a 110W-h power bank (30,000 mAh). To reduce the power consumption, some components, such as High-Definition Multimedia Interface (HDMI) port, Bluetooth, and activity Light Emitting Diode (LEDs) were deactivated (Droissart et al., 2021).

Communicating the information from the camera trap like devices can help to protect the ecosystems with timely data availability (Whytock et al., 2023). The species classification used Google AutoML on an AI enabled camera trap and RPi was used to transmit the alerts comprising image labels on the Iridium satellite network. The system was shown to run for 3 months without any downtime (Whytock et al., 2023).

In contrast to these studies, the novelty of this study is in developing a continuously operating prototype camera trap platform based on the low-power RPi-Z device for on-board image classification with fine-tuned DNN models, and visualizations with the XAI techniques.

2.2. Transfer learning with fine-tuning of small models

The DNN models like MobileNetV2 have a small size, better performance, and processing requirements that can be fulfilled on an edge device in the field. Dufourq et al. (2022) conducted a comprehensive investigation of feature extraction approaches for PAM using 12 pre-trained CNN models including MobileNetV2 with a classifier layer added, demonstrating good performance with CPU processing. The authors also suggested use of fine-tuning approaches, that is training of last few layers, after sufficient data becomes available (Dufourq et al., 2022). MobileNetV2, EfficientNetB0, and EfficientNetB3 models were used for animal classification on the Raspberry Pi with a focus to analyse the trade-off between the model precision and inference. The images from Caltech camera trap dataset and Snapshot Serengeti dataset were used (Cunha et al., 2021).

In comparison to the studies covered in this section, the novelty of this study is in highlighting the model performance improvement with fine tuning (that is, training some or all the layers of a pre-trained model) of the selected DNN models, and hyperparameter optimization with two techniques. These transfer learning approaches have not been fully explored in ecological studies, and can not only improve the classification results, but also provide trade-offs for model size and accuracy versus available time and compute resources. To the best of our knowledge, this is the first study proposing the use of transfer learning with fine tuning to improve the classification results, and the use of vision transformer (ViT) architecture for image classification.

2.3. XAI techniques

An ensemble of five DNN models and XAI techniques were used for determining and understanding seagrass distribution and the environmental variables (He et al., 2022). The XAI techniques were found to improve the understanding of the species distribution models that are used for estimating and predicting the species distribution in time and space (Ryo et al., 2021). The use of XAI techniques were demonstrated for distribution modelling of African elephant (*Loxodonta Africana*) with Random Forest algorithm to evaluate model reliability and the importance of the predictor variables, important for understanding the spatial context. It was concluded that by combining the XAI techniques with the biogeographical and ecological knowledge, better model interpretations were possible (Ryo et al., 2021). The impact of visualization techniques for human-AI trust were studied for morphological clarity (MC) using Grad-CAM (Karran et al., 2022).

In contrast to the studies covered in this section, the novelty of this

study is in providing a comparison of two XAI techniques for image visualization with two DNN models. These visualization techniques have not been explored before for image understanding on-board a camera trap for animal classification ecological studies.

3. Materials and methods

In this section, we describe the datasets, and the overall process flow used in this study for image classification and visualization. For image classification, we provide a comparison of three different ways in which the transfer learning techniques can be used for a trained model, suitable for deployment to a RPi-Z device. We briefly describe the various models used for image classification, hyperparameter optimization, and the performance metrics. For image visualizations, we describe the two selected XAI techniques.

3.1. Image datasets

3.1.1. Danger of extinction dataset

This dataset comprises of eleven animal species, with a total of 6484 images (Extinction dataset: Danger of Extinction / Animal Image Set, n.d.), as shown in Table 2. It is a balanced dataset of the selected species under danger of extinction according to International Union for Conservation of Nature (IUCN) Red List Of Threatened Species. The images are in different resolutions and were scaled for the different DNN models used in this paper. The dataset comprises of images with a clear view of an animal, however some images included in the dataset are quite challenging. Some images from the dataset are shown in Fig. 1.

3.1.2. Florida wildlife camera trap dataset

This dataset comprises of a total of 104,495 images collected from two locations in Southwestern Florida (Gagne et al., 2021). It is a challenging dataset for image classification as the images from the field deployments are often affected by the environmental factors, such as illumination issues, animal proximity to the camera, occlusion, motion blur. We chose a subset of the dataset, made available as Part1.zip (Gagne et al., 2021), comprising a total of 12,706 images of 13 animal species (Table 3), and refer to this subset as Florida-Part1 dataset. Some of the sample images are shown in Fig. 2.

3.2. Process flow

This study has two perspectives, model training and development on a high-end computer, and the model deployment to an edge device, as shown in Fig. 3. The training dataset could be pre-processed, such as denoising, contrast enhancement, and image resizing. We only used image re-sizing to make the images suitable for the input image resolution required for the selected models.

After a model has been trained, it needs to be validated and tested to confirm that it has achieved the desired performance. The performance might be measured using different metrics, such as classification

accuracy. For improving the model performance, the training process can be repeated by changing to another set of the model's hyper-parameters. After the desired performance has been achieved, the trained model can be used for making inferences or predictions.

For model deployment to an edge device, e.g., RPi, the model needs to be converted to the TensorFlow Lite format, which although reduces its size, but also reduces the model's accuracy. The model can then be used on an edge device for classification of as yet unseen images. The TensorFlow or TensorFlow Lite model may also optionally be optimized for size, or latency depending on the application requirements and the edge device specifications. The RPi or similar edge devices can also be used to run XAI techniques such as Grad-CAM for generating visualizations through heatmaps.

3.3. Transfer learning approaches for image classification

Image classification models can have many layers and millions of model parameters to be learned on a training dataset. This requires lots of training data, processing power, and can take a long time. Transfer learning means that a model trained on one dataset, can be used for image classification on another dataset with little or no changes. Transfer learning utilizes a pre-trained model M_{pt} trained on a large original dataset D_{orig} with number of classes C_{orig} , for image classification on another dataset D_{new} with number of classes C_{new} .

This transfer of the knowledge or learning from the original dataset D_{orig} to dataset D_{new} is possible because the initial and the middle layers of a DNN during training learn the low and medium complexity features in an image, such as edges, whereas the layers towards the end of the model represent high level features based on the training dataset. The transfer of model's learning or knowledge can thus be exploited for a similar task. We describe below the three common transfer learning approaches as shown in Fig. 4.

3.3.1. Direct use of a pre-trained model

The direct use of a pre-trained model means that a pre-trained model trained on large datasets such as ImageNet (ImageNet Data, n.d.), or COCO (Lin et al., 2014) is used for classification of images from another similar dataset. This approach was used to classify animals (Carl et al., 2020; Duggan et al., 2021) and although has the advantage that no model training is required but the classification accuracy will be low. A total of 36 models pre-trained on ImageNet, were used without any additional retraining of the model with the new camera trap images, and it was found that the results were poor as the camera trap images were different and challenging compared to the ImageNet images (Stancic et al., 2022). Although the pre-trained model can be used as such for classification, but commonly, a classifier head is added to make the pre-trained model suitable for the number of classes in the new dataset. For further model performance improvement, some or all the layers in the pre-trained model can be re-trained. Therefore, there is a need for some adaptation of a pre-trained model to achieve better performance, as described in the sections 3.3.2 and 3.3.3.

The pre-trained models are also useful where the data labelling is costly, or the datasets are small. There is a trade-off between the model size and accuracy with larger models tending to be more accurate but could result in processing latency on edge devices, such as RPi.

3.3.2. Transfer learning - feature extraction

In order to use a pre-trained model for feature extraction, the features already learned by the pre-trained model are retained by adding a classification layer or classifier (based on the classes in the new dataset). Thus, the pre-trained model acts as a feature extractor for the new model with the added classifier. For example, a pre-trained model trained on ImageNet dataset will have a classifier comprising of 1000 classes, which can be replaced by a classifier with 13 classes for the Florida-Part1 dataset. This technique was used for image classification on

Table 2
The class distribution in the Extinction dataset.

Class	Number of images
African_Elephant (Loxodonta)	470
Amur_Leopard (Panthera pardus orientalis)	530
Arctic_Fox (Vulpes lagopus)	369
Chimpanzee (Pan troglodytes)	462
Jaguars (Panthera onca)	577
Lion (Panthera leo)	806
Orangutan (Pongo pygmaeus)	496
Panda (Ailuropoda melanoleuca)	846
Panthers (Panthera onca)	565
Rhino (Diceros bicornis)	773
Cheetahs (Acinonyx jubatus)	590

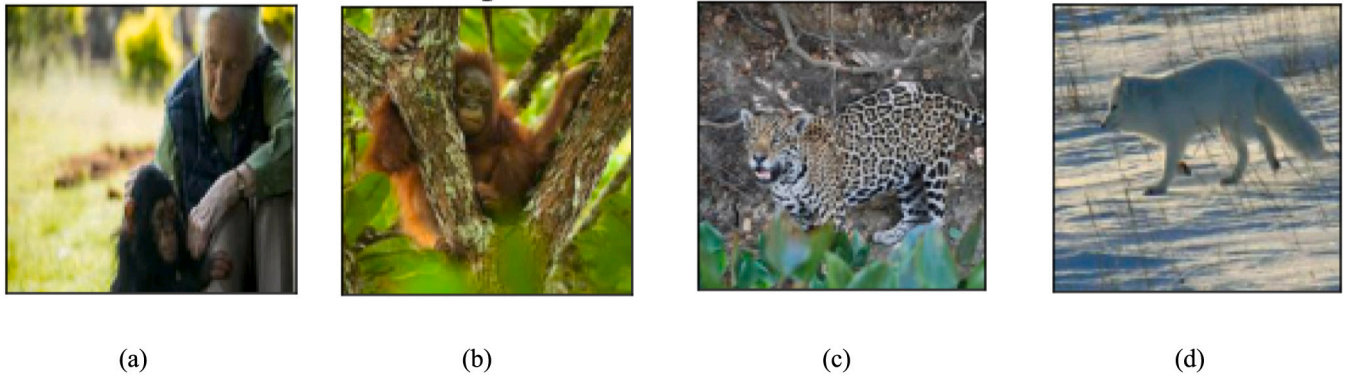


Fig. 1. Sample images from the Danger of Extinction dataset (a) Chimpanzee (*Pan troglodytes*) (b) Orangutan (*Pongo abelii*) (c) Jaguar (*Panthera onca*) (d) Arctic Fox (*Vulpes lagopus*).

Table 3

The class distribution in the Florida-Part1 dataset.

Class	Number of images
Alligators (<i>Alligator mississippiensis</i>)	27
Armadillos (<i>Dasypodidae</i>)	130
Bears (<i>Ursidae</i>)	512
Birds_Carrion	1346
Birds_Other	2450
Birds_Raptors	73
Birds_Wading	3821
Boar (<i>Sus scrofa</i>)	117
Bobcats (<i>Lynx rufus</i>)	2463
Coyotes (<i>Canis latrans</i>)	146
Dogs (<i>Canis lupus familiaris</i>)	196
Opossum (<i>Didelphis Virginiana</i>)	1401
Otters (<i>Lutra Canadensis</i>)	24

dataset labelled by the citizen scientists on Zooniverse (Willi et al., 2019), and Serengeti dataset (Villa et al., 2017). This is the most common use of the transfer learning approaches as this requires the least knowledge of the model, and is easy and quick to implement (Dufourq et al., 2022). The use of fine-tuning approaches, that is training of last few layers, was suggested if more data were available (Dufourq et al., 2022).

3.3.3. Transfer learning – Fine tuning

The idea of fine tuning with a pre-trained model, could be used to train some or all of the layers of the pre-trained model, whereas the initial model layers could be used as such, thus reducing the overall time to train the model. The benefit is that the low-level features learnt by the initial layers, which are common to any classification problem are

utilized, whereas the later layers (to be fine-tuned) learn the high-level features of the new dataset during the model training.

Training the last few layers of a pre-trained model helps the model to learn the features in the new dataset and help to improve the model accuracy on the new task. In general, with fine-tuning more layers the model performance will improve as more of the pre-trained model parameters will be updated or tuned to the new dataset, but this can have other considerations, such as the new dataset size, and its similarity to the original dataset. Fig. 5 shows the different ways in which feature extraction or transfer learning approaches will suit a new dataset based on its size and similarity to the original dataset. This is included for guidance only, and the actual adoption of a technique should also consider the particular classification problem. It was found that the transfer learning worked well if the dataset was small (Willi et al., 2019; Norouzzadeh et al., 2018).

3.4. DNN models

We used the Python programming language for the application development. We used TensorFlow and Keras to implement the DNN models. The DNN models were trained and tested on Google Colaboratory (Google Colab, n.d.), a free browser-based online environment with Graphics Processing Unit (GPU) access, and were then deployed on the RPi-Z device.

3.4.1. MobileNetV2

The MobileNet models were specifically designed for low power devices, such as mobile phones and edge devices like RPi-Z. The MobileNet model (Howard et al., 2017) has the least size and low latency compared to other DNN models (Keras, n.d). Although these models do not have the highest accuracy (Table 4), but the size, performance, and

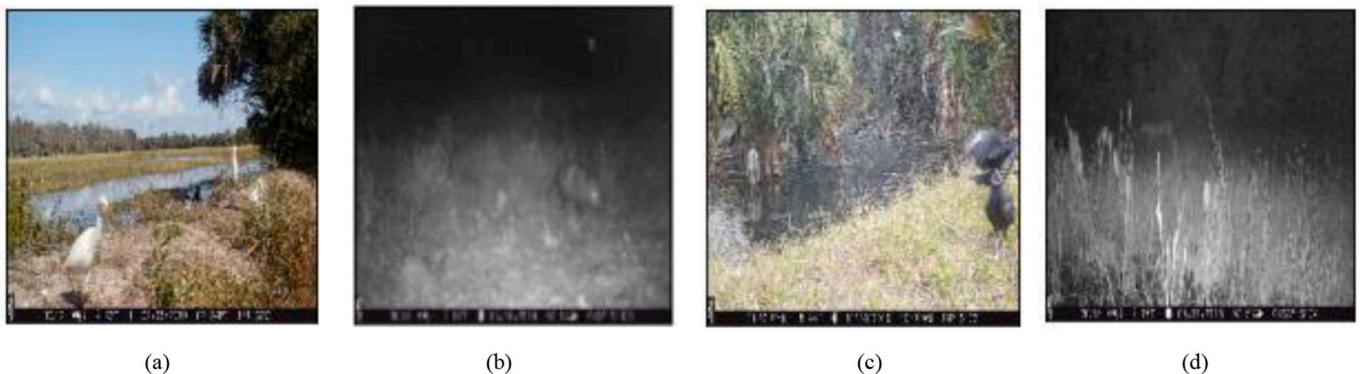


Fig. 2. Sample images from the four classes from the Florida-Part1 dataset (a) Birds_Wading (b) Opossums (*Didelphis Virginiana*) (c) Birds_Carrion (d) Bobcats (*Lynx rufus*).

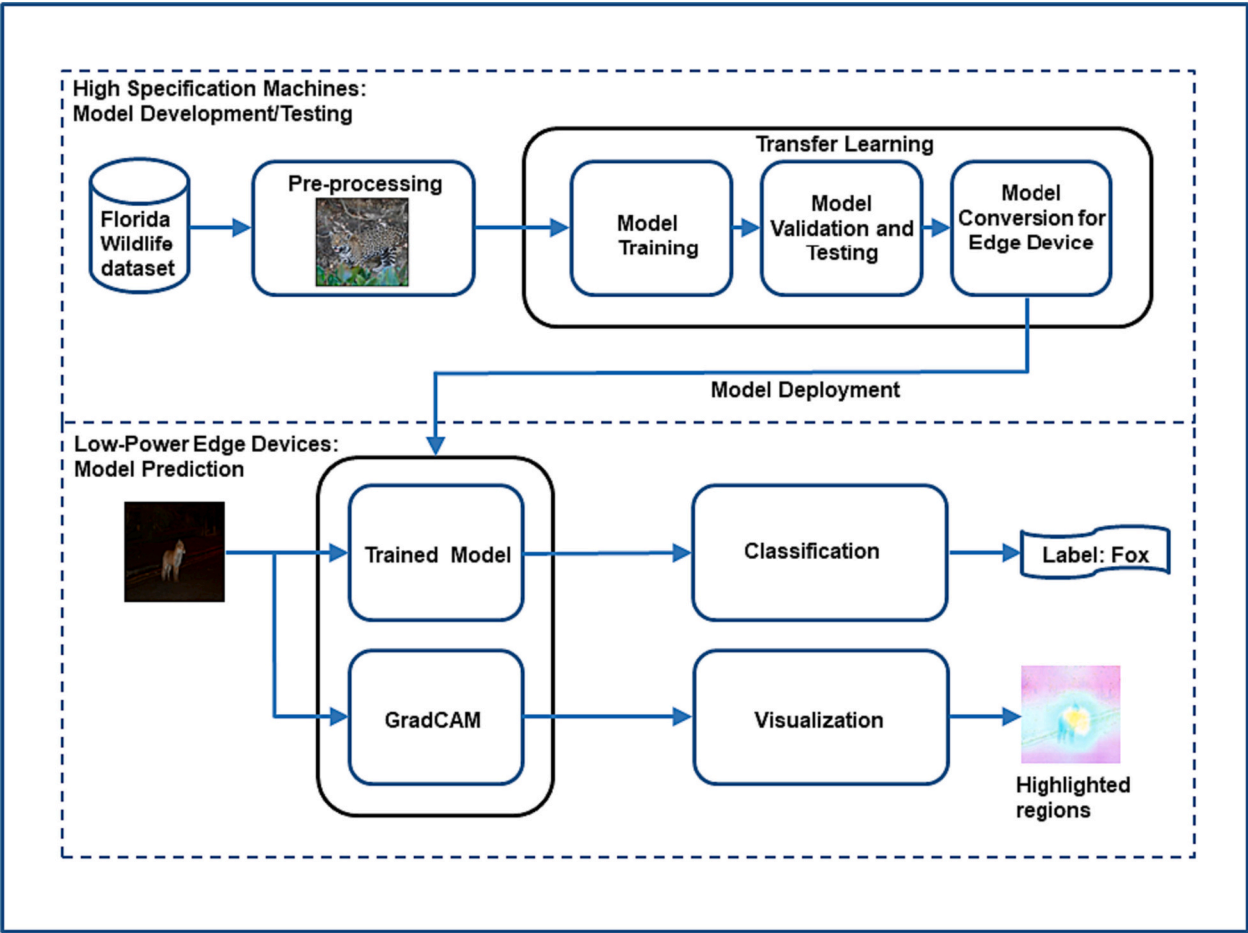


Fig. 3. The process flow of image classification and visualization.

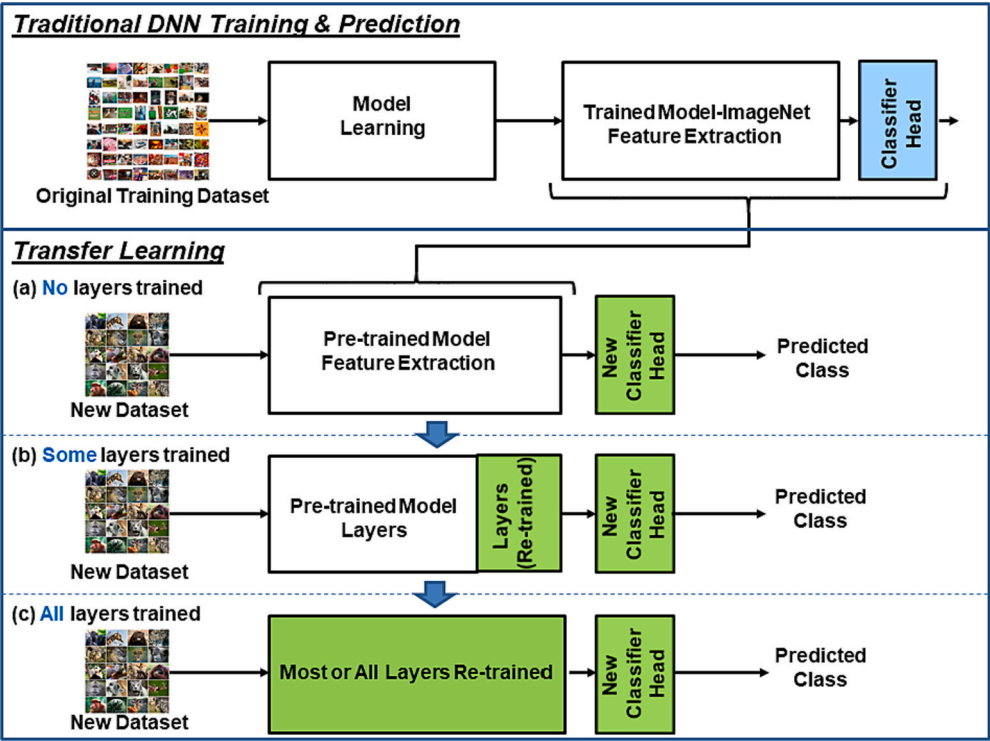


Fig. 4. Transfer Learning approaches.

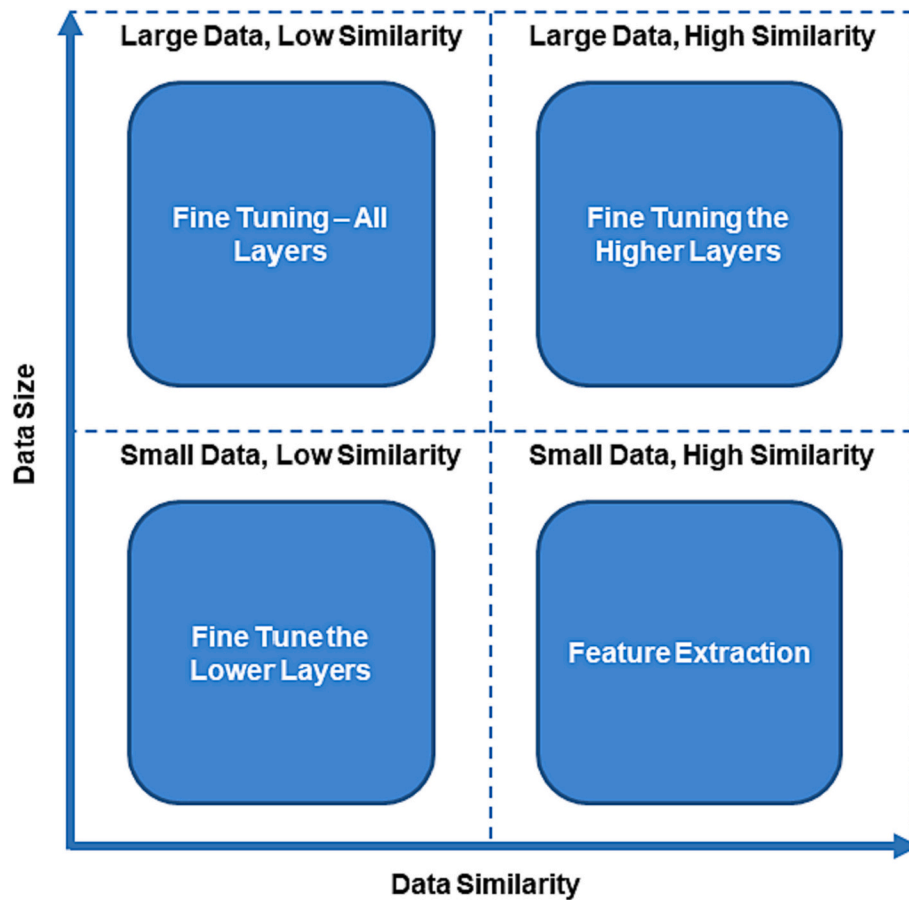


Fig. 5. The applicability of the Transfer Learning approaches based on the new dataset size and similarity to the original dataset.

latency advantages make MobileNet models an ideal choice for image classification on edge devices. The model uses simple convolution operations for a light-weight DNN. The trade-off between latency and accuracy was made possible through the use of two model hyperparameters (Howard et al., 2017). A new mobile architecture, MobileNetV2 for edge applications was proposed with a relatively smaller size in comparison to the other models as shown in Table 4. The smaller size of the model was achieved by depth separable convolutions with residuals (Sandler et al., 2018). The characteristics of the MobileNet models along with VGG (Visual Geometry Group)-19 model are

Table 4
Size comparison of the trained models (Keras, n.d).

Model	Size (MB)	Top-1 Accuracy (%) on ImageNet-1 k	Model Parameters	Reference
MobileNet	16	70.4	4.3 M	(Howard et al., 2017)
MobileNetV2	14	71.3	3.5 M	(Sandler et al., 2018)
MobileNetV3	-	75.6	5.4 M	(Howard et al., 2019)
EfficientNetV2B0	29	78.7%	7.2 M	(Tan and Le, 2021)
MobileViT-XXS	-	69%	1.9 M	(Mehta and Rastegari, 2022)
MobileViT-S	-	78.3%	5.7 M	(Mehta and Rastegari, 2022)
VGG-19	549	71.3	143.7 M	(Simonyan and Zisserman, 2015)

provided in Table 4.

We used transfer learning in three possible ways by using a model pre-trained on the ImageNet dataset (ImageNet Data, n.d) with 1000 image classes. The pre-trained MobileNetV2 model was used for feature extraction by adding a classifier head (Fig. 4). The MobileNetV2 model was thus used for classification for 11 and 13 classes in the Extinction and Florida-Part1 dataset. We refer to these models as FeatureExtraction in the rest of this paper.

The MobileNetV2 base model comprises of 154 layers, and we used fine tuning to train the model in two ways, by training only the last 20 layers and all layers, to create two different models. Please note that in the layer count, we are considering all the model layers, both trainable and non-trainable. The selection of 20 layers was based on aligning the selection with the last block boundary in the MobileNetV2 architecture as shown in Fig. 6.

Fine tuning requires a low learning rate, and we used a learning rate of 0.00001 determined through hyperparameter optimization (Section 3.5). For simplification, we refer to the model fine-tuned with 20 layers as FineTuned-20, and the other model with fine tuning of all layers as FineTuned-All in the rest of this paper.

We used the pre-trained MobileNetV2 model that has an architecture as shown in Fig. 6. The input images were normalised, and resized to 224×224 pixels. The Extinction dataset was split into 80% training, 10% validation, and 10% test images, comprising 5182, 644, and 656 images belonging to 11 classes, Florida-Part1 dataset was split into 70% training, 10% validation, and 20% test images, comprising 8889, 1266, and 2551 images respectively belonging to 13 classes (Gagne et al., 2021). For all the models' training, a batch size of 32, with 200 epochs, and an 'Adam' optimizer were used.

The trained model can be ported using TensorFlow Lite (a version for

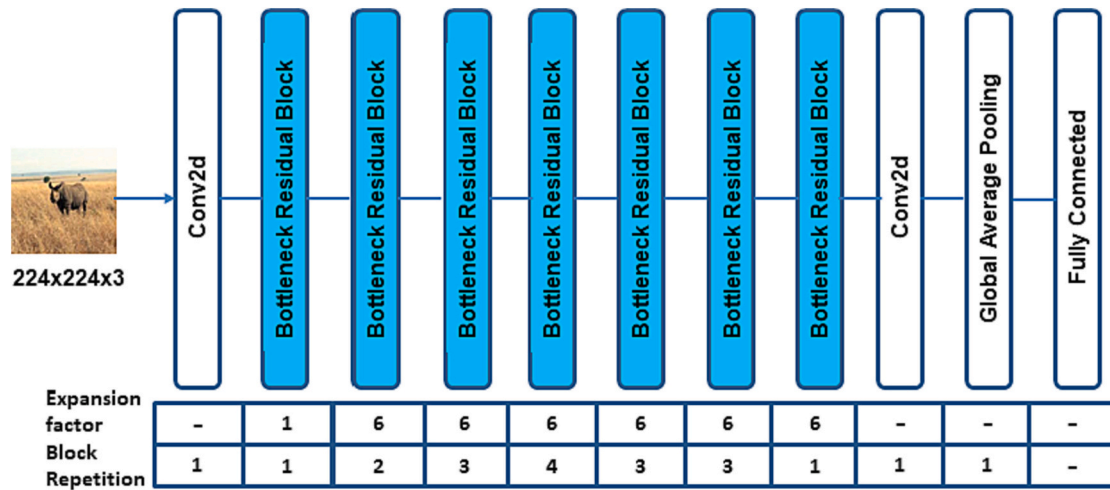


Fig. 6. The layered architecture of the MobileNetV2 model.

low power devices) for deployment to an edge device for image classification on-board the edge device.

3.4.2. Model comparison

The selection of a model for an embedded application will be governed by the application latency needs, model size, and accuracy, providing different selection trade-offs according to the requirements. In this paper, we have chosen to use MobileNetV2 due to its small size and better performance, and due to its use in many studies (Cunha et al., 2021; Kuncheva et al., 2023; Yang et al., 2022). For performance comparison, we have also considered few other models that although can outperform MobileNetV2 in terms of accuracy and latency, but not size. We provide a brief description of the selected models here and more details can be found in the references provided.

Neural architecture search (NAS) was used for developing the EfficientNetV2B0 model, and for the optimization of parameter efficiency and training speed (Tan and Le, 2021). The model performance was improved using progressive learning to adapt regularization to the image size (Tan and Le, 2021). EfficientNetV2 is a family of models (B0 to B7), and B0 is the smallest variant in the family (Keras, n.d), with the other models progressively increasing in performance going up to EfficientNetV2B7.

Vision Transformers (ViT) (Dosovitskiy et al., 2021) are based on an alternative attention-based model design rather than CNN. The transformer architecture was first applied on Natural Language Processing (NLP) problems. The recent ground-breaking research outputs of Generative Pre-trained Transformer (GPT) and Bidirectional Encoder Representations from Transformers (BERT) are all based on the transformer architecture. The concepts of attention were later applied to the image classification problems as Vision Transformers (Dosovitskiy et al., 2021). The transformers use the relationship between words, termed as attention, whereas the vision transformers determine relationships between patches of images. ViT have since achieved performance similar to convolution-based architectures. There are many vision transformer architectures for image classification but we chose MobileViT (Mehta and Rastegari, 2022) models proposed as a small model for embedded applications (TensorFlow Hub, n.d). MobileViT-S was shown to perform better than MobileNetV2 and MobileNetV3 (Mehta and Rastegari, 2022).

There are three variants of the MobileViT models from smallest to the largest, MobileViT-XXS, MobileViT-XS, and MobileViT-S pre-trained on the ImageNet-1 k validation set, with top-1 accuracy as 69.0, 74.7, and 78.3 respectively (TensorFlow Hub, n.d). An implementation can use any of these transformer models depending on the size and accuracy requirements.

We used MobileViT models as a feature extractor, that is, adding a classifier head, and also by training all the model layers, referred to as FineTuned-All.

3.5. Hyperparameter optimization

The model training process determines the values for the trainable parameters of a model, e.g., Table 4 shows the size of the model parameters. In addition, a DNN model also has other parameters that need to be selected, e.g., batch size, that determines the model performance. The model parameters define the model and are termed as hyperparameters. Each hyperparameter has a range of values, from which an optimum selection can improve the model performance. The model hyperparameters could be in hundreds, presenting a very large search space, but these differ in their relative importance. The common ones for a DNN are learning rate, batch size, and dropout.

Hyperparameter optimization techniques make it simpler to search for an optimum hyperparameter configuration, however, it can take a significant amount of time depending on the number of hyperparameter values being searched (Nazir et al., 2020a). We selected the Keras Tuner (O'Malley et al., 2019) which provides Random Search, Hyperband, and Bayesian Optimization algorithms. In this paper, we used Hyperband and Bayesian Optimization as these two algorithms are superior to the other search algorithms available in the Keras Tuner.

To illustrate the effect of a correct hyperparameter choice, we chose to optimize the learning rate for the MobileNetV2 and EfficientNetV2B0 using Hyperband and Bayesian Optimization. This is an important hyperparameter associated with the DNN, and a low value of the learning rate means that although the model would try to converge quickly, it may settle for an erroneous value, and similarly a higher than required value would not only take a long time, but would also not result in an optimal value.

3.6. Image visualization

The DNN models are opaque, much like 'black-boxes', with a lot of parameters and layer interconnections. Although the prediction performance is high, it is difficult to explain why and how a particular prediction was made. Explainable AI (XAI) techniques for image classification, can help a human expert, for example, an ecologist to understand the decisions made by a trained model. Fig. 7 shows the need and context for explainability in understanding the model's predictions.

The heatmap in Fig. 7, can be generated using different colormaps (OpenCV, n.d), such as the one shown in Fig. 8. The importance of an image region for the model's prediction, is represented with a color from

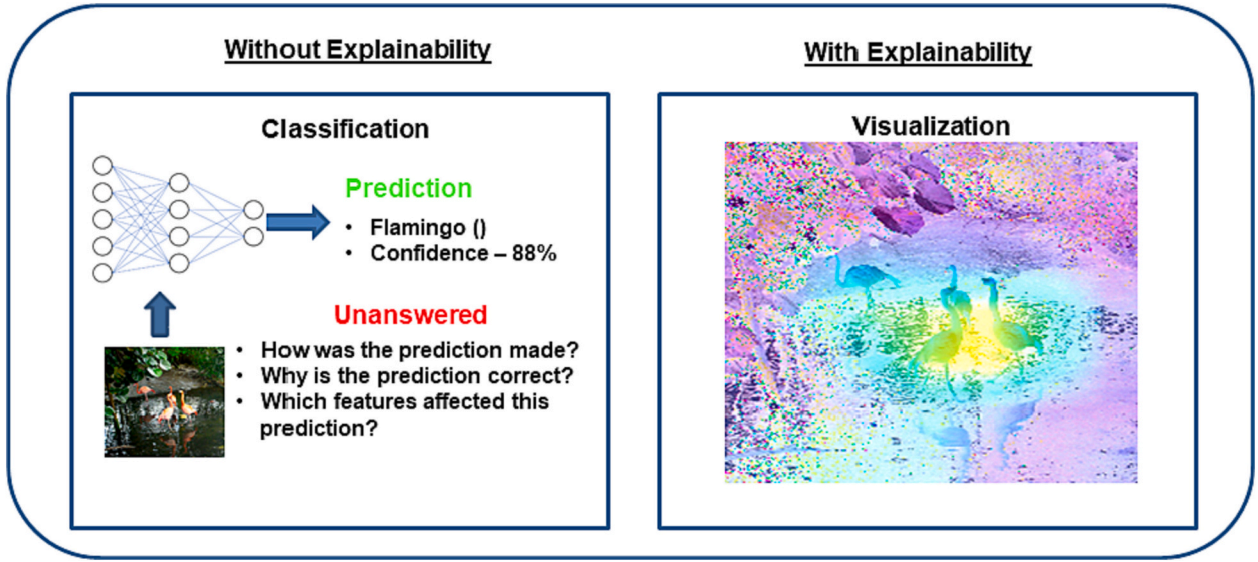


Fig. 7. Explainability of the model's decision highlighting the important image regions for the model prediction.

the colormap, with a progressively increasing importance (left to right), with yellow representing the most important regions.

The visualizations can help to answer the questions as to which areas in the image contributed to the particular classification prediction, and if the DNN model is focussing on some background features for its decision, such as, attributing snowy terrain to predict an arctic fox.

The aim of the XAI techniques is to provide additional information to explain the decision for the predicted class, which could be textual information, similar selected images for the predicted class, or image highlighting using heatmaps to indicate the image areas or features considered significant by the DNN for its classification decision. We used tf-explain (tf-explain, n.d) to determine the visualizations with heatmaps for the two techniques, Grad-CAM (Selvaraju et al., 2017) and Occlusion Sensitivity (Zeiler and Fergus, 2014). Some of the model visualization techniques need the model to be re-trained, whereas techniques, such as Grad-CAM and Occlusion Sensitivity can be used without re-training the models.

Grad-CAM is a popular visualization technique and can be applied to a wide variety of DNN models (Selvaraju et al., 2017). Grad-CAM can be used to find the regions in the image which were considered significant by the DNN for making its prediction. Grad-CAM visualizes the image regions that are significant for the classifier's prediction by using the activation maps (tf-explain, n.d). The mathematical relationships for the calculation of the Grad-CAM are provided below and the full details of the method can be seen in the paper (Selvaraju et al., 2017).

For a class c , the gradient y^c with respect to the feature map activations, A^k is obtained as shown in eq. (1) (Selvaraju et al., 2017).

$$\text{Gradients} = \frac{\partial y^c}{\partial A_{ij}^k} \quad (1)$$

The gradients are average pooled (over the width and height) to find the neuron importance weights, α_k^c , representing a partial linearization of the downstream network, as shown in eq. (2) (Selvaraju et al., 2017).

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \text{gradients} \quad (2)$$

A weighted combination of the forward activation maps is given by eq. (3) (Selvaraju et al., 2017).

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (3)$$

This creates heat map of the input image highlighting the significant regions of the image using the gradient information of the final convolutional layer (Selvaraju et al., 2017).

In Occlusion Sensitivity, the visualization is based on a multi-layered Deconvolutional Network (deconvnet) used to project the feature activations to the input pixels, to determine the input patterns causing the feature map activations. A deconvnet can be considered as a convnet but with the layer operations reversed. The importance of image areas important for prediction are determined by systematically hiding the image areas, and observing the effect on the classifier's prediction (Zeiler and Fergus, 2014). If the image areas having significance for the classifier's predictions is occluded then the probability of the correct class will also drop considerably.

The different contiguous rectangular patches of the input image, $p_i \in P$, are replaced with the given patch (such as, all-zero) and the decrease in prediction features $f_c(x)$ are monitored (Kakogeorgiou and Karantzas, 2021; Zeiler and Fergus, 2014).

$$\mathcal{O}_{\text{Occlusion}}^{p_i}(f_c, x) = f_c(x) - f_c(x_{p_i}) \quad (4)$$

In this paper, we used the implementation of the Occlusion Sensitivity (tf-explain, n.d) with a patch size of 16 that determined the size of the region to be occluded.

3.7. Edge device – Camera trap prototype

We used RPi-Z as the camera trap device to run the trained DNN model. RPi-Z is a microcomputer device with 32-bit ARMv6Z architecture and General-Purpose Input/Output (GPIO) pins for interfacing sensors, such as Passive Infrared (PIR) sensor, making it an ideal choice for prototyping. The device has a connection slot for an on-board camera with 5 Megapixels (2592 × 1944) resolution. The device can run many variants of Linux Operating System (OS) with a removable SD card used as bootable and storage device. A recent higher specification version,



Fig. 8. Colormap - COLORMAP_VIRDIS.

RPi 4, was found to consume the least current in comparison to the other selected devices (Zualkernan et al., 2022). However, compared to RPi 4, RPi-Z has much lower power consumption, making it an ideal choice for long-term battery-powered applications.

We used a chargeable Universal Serial Bus (USB) power supply designed for powering mobile phones, rated at 900000 mAh to power the RPi-Z. The power supply can be charged using 5 Volts supply, and additionally has a solar panel, making it ideal for continuous operation for long term deployments.

For image classification on the edge devices, the TensorFlow Lite and TensorFlow Lite runtime is required on the edge device.

3.8. Performance metrics

The metrics commonly used to evaluate the DNN classification performance on the test data are derived based on the relative numbers of the True Positive (TP), True Negative (TN), False Negative (FN), and False Positive (FP) images. In this paper, we have used the performance metrics of accuracy, precision, recall, and F1-score (Scikit-learn, n.d) as described below:

Accuracy is the ratio of correctly predicted instances and total instances but may not be reliable with imbalanced classes. The accuracy is defined as

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP} \quad (5)$$

Precision is the ratio of correctly predicted positive instances and the total positive instances, and is defined as

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

Recall indicates the number of positive instances that are classified correctly, and is defined as

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

F1-score indicates the harmonic mean of precision and recall and is defined as

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (8)$$

The problem under consideration in this research is multi-class classification, therefore in later sections we have reported both the macro and weighted measures of the performance metrics. These measures are helpful to understand a model's performance in multi-class classification. In eq. (9) and (10) we have shown the Macro Precision and Weighted Precision, but this is applicable to all the metrics in eq. (5) to (8).

Let C_i to C_N be the N classes, with the number of occurrences of each class in the dataset as W_i to W_N , and the Precision of each class as P_i to P_N , then the Macro Precision and Weighted Precision can be represented as:

$$Macro\ Precision = \frac{\sum_{i=1}^N P_i}{N} \quad (9)$$

$$Weighted\ Precision = \frac{\sum_{i=1}^N P_i W_i}{N} \quad (10)$$

4. Results

4.1. Image classification with MobileNetV2 model

The model MobileNetV2 was trained and tested on the Google Colab (Google Colab, n.d) for the two selected datasets. The model

classification accuracy results are shown in Table 5. The training accuracy is generally higher than the validation and testing accuracies. The training, validation and testing accuracies in Table 5 can be seen to increase for both datasets in progressing from the FeatureExtraction to FineTuned-All. The performance of FeatureExtraction model is lower compared to the fine-tuned models, because it was pre-trained on ImageNet dataset but is being used to classify images from the Extinction and Florida-Part1 datasets.

The training and validation accuracy for the Extinction dataset with the three models are shown in Fig. 9. The improvement in the model's accuracy can be seen across Fig. 9 (a) to (c). It can be seen in Fig. 9 (a) that the validation accuracy starts at a high initial value because in FeatureExtraction we are using the features that this model had learnt during training on the ImageNet dataset. The models were trained for 200 epochs and it can be seen that for the FineTuned models, better model accuracy is achieved in comparison to the FeatureExtraction. The improvements in the FineTuned models' plots are also relatively smoother compared to FeatureExtraction due to its low learning rate. The graphs for the Florida-Part1 dataset were similar and are not shown.

The performance metrics for the two datasets extracted from the classification report (Scikit-learn, n.d) are provided in Table 6. These results show both the macro and weighted averages for the performance metrics. The macro average values are the simple averages of e.g., precision across all classes in the dataset, whereas the weighted averages also account for the number of images in each class.

The macro and weighted average values are same for the Extinction dataset due to its balanced data. However, the differences between the macro and weighted values can be seen across the Florida-Part1 dataset which is an unbalanced dataset.

For the Florida-Part1 dataset, we gathered the statistics using the confusion matrix (Scikit-learn, n.d) to derive the performance metrics for each class as shown in Table A1 (Appendix A). There were a total of 2551 test images with a distribution across the thirteen classes as shown by the numbers in the first row of Table A1, e.g., the total images for the class 'Alligators' is shown as 7, and this number can also be obtained by adding the TP and FN values for this class. The results for the accuracy, precision, recall, and F1-Score for the Florida-Part1 dataset are provided in Fig. 10.

It can be seen from Fig. 10, that there is a performance improvement in going from FeatureExtraction (AccuracyFE), to FineTuned-20 (Accuracy-20), and then to FineTuned-All. Significant improvements can be seen across the class 'Alligators', 'Boars', 'Coyotes', and 'Otters', and from Table A1 it can be seen that these classes have fewer image samples present. However, for 'Birds-Raptors', there is some improvement, but the values indicate that the images were not classified correctly. For this class, the low values for precision, recall, and F1-Score indicate the possible class overlap with the other three birds' classes, despite this, the accuracy value is high which can often be misleading as a performance metric.

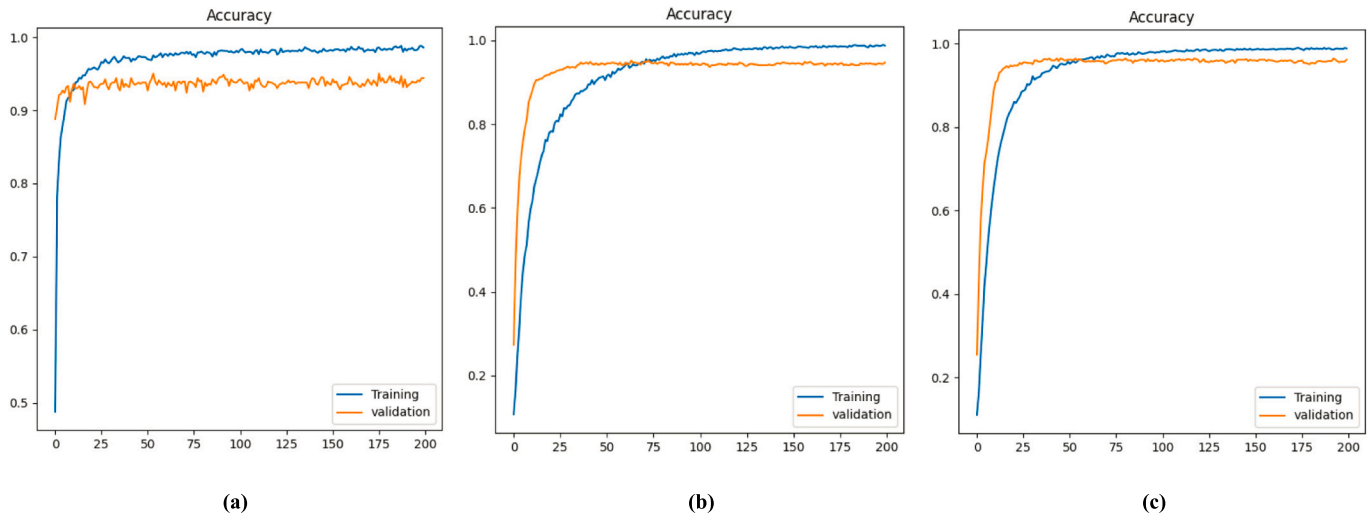
4.2. Image classification on the edge device

The image classification on the RPi-Z device used the TensorFlow Lite version of MobileNetV2 model, FineTuned-All trained on the Florida-Part1 dataset. As the model was trained on Florida-Part1 dataset, therefore six images with different characteristics were selected from the Florida-Part1 dataset for classification on the RPi-Z. The results for the six selected images from the Florida-Part1 dataset are shown in Fig. 11. The prediction of the DNN model is correct for the images that have a clear view and size of the animal. However, the images 11 (c)-(f) can be challenging even for a human expert. The only image classified incorrectly is shown in Fig. 11(d), which is an otter but classified as coyote. However, this is correctly classified considering Top-5 accuracy, often reported in classification tasks (Norouzzadeh et al., 2018; Villa et al., 2017).

Table 5

Accuracy of the selected models on the datasets.

Dataset	Classification Accuracy								
	FeatureExtraction			FineTuned-20			FineTuned-All		
	Training	Validation	Testing	Training	Validation	Testing	Training	Validation	Testing
Extinction	0.9859	0.9441	0.93	0.9878	0.9472	0.96	0.9884	0.9612	0.96
Florida-Part1	0.9010	0.9068	0.89	0.9912	0.9613	0.96	0.9921	0.9629	0.96

**Fig. 9.** Training and Validation Accuracy for the Extinction dataset (a) FeatureExtraction (b) FineTuned-20 (c) FineTuned-All.**Table 6**

Performance of the selected models on the datasets.

Transfer learning approach	Performance metric (average values)	Extinction Dataset		Florida-Part1 Dataset	
		Macro Average	Weighted Average	Macro Average	Weighted Average
FeatureExtraction	Precision	0.93	0.93	0.54	0.87
	Recall	0.93	0.93	0.52	0.89
	F1-score	0.93	0.93	0.53	0.88
FineTuned-20	Precision	0.96	0.96	0.90	0.96
	Recall	0.96	0.96	0.84	0.96
	F1-score	0.96	0.96	0.86	0.96
FineTuned-All	Precision	0.97	0.97	0.88	0.95
	Recall	0.97	0.96	0.87	0.96
	F1-score	0.97	0.96	0.87	0.95

4.3. Model performance comparison

The performance comparison of MobileNetV2, EfficientNetV2B0, MobileViT-XXS, and MobileViT-XS models with all layers fine-tuned, for image classification on the Extinction dataset is shown in Fig. 12. It can be seen that the performance of the EfficientNetV2B0 model is better than the other models. As can be seen, the MobileViT-S model showed better performance compared to the smaller model MobileViT-XXS. MobileViT-S model based on the vision transformer also achieved performance better than the convolution-based architecture, MobileNetV2.

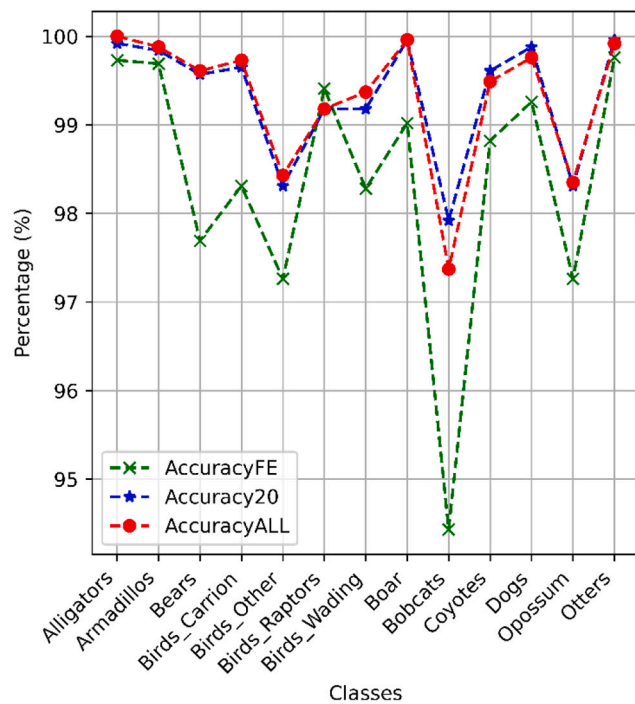
4.4. Hyperparameter optimization

The hyperparameter optimization results for the learning rate for the EfficientNetV2B0 and MobileNetV2 model were obtained using the Hyperband and Bayesian Optimization for the Extinction dataset. The optimum values determined using the Keras Tuner for the learning rate were 0.00001 for both the models. The F1-Scores were then obtained for the optimal learning rate (0.00001), a higher value (0.0001), and a lower value (0.000001) and are shown in Fig. 13. It can be seen that the

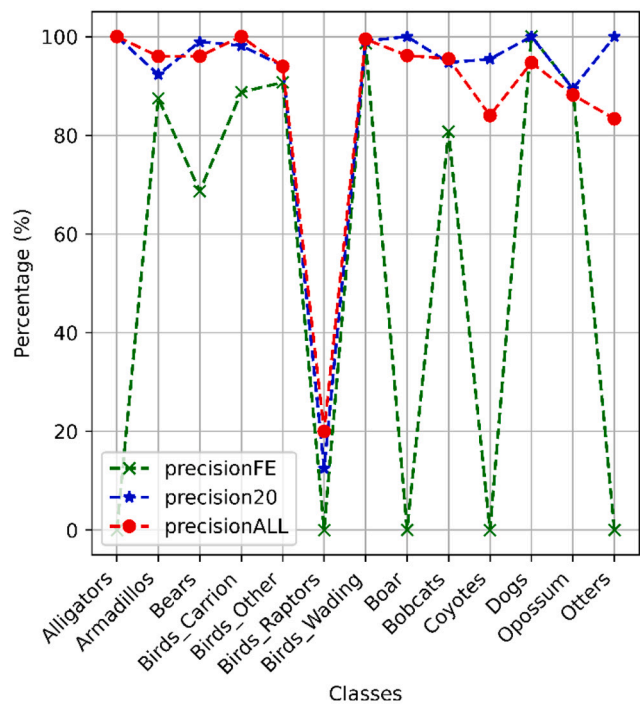
optimal learning rate found through the hyperparameter optimization provides better model classification performance for the selected models and the dataset.

4.5. Visualising predictions

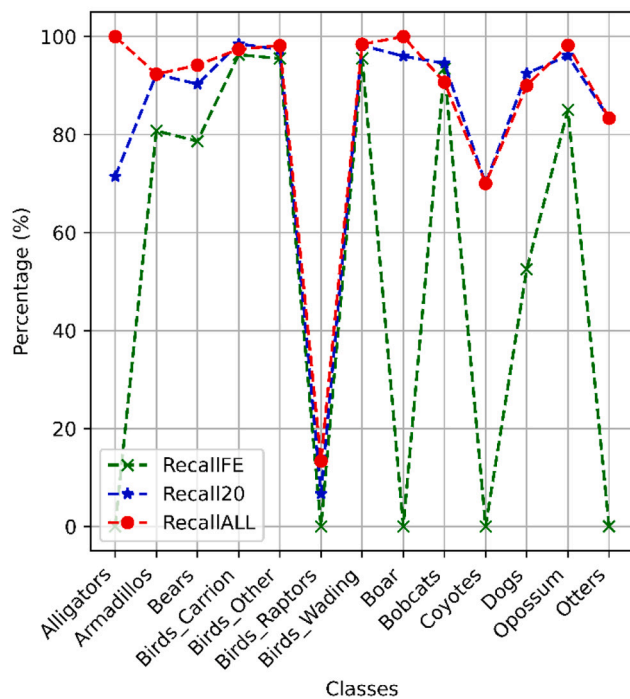
The Grad-CAM and Occlusion Sensitivity visualization results obtained with VGG-19 and MobileNetV2 model on images from the ImageNet dataset are provided in Fig. 14. The colors assigned to various regions are from the colormap shown in Fig. 8 with the regions shown in yellow being significant for the model's classification decision, and a progressively lesser significance shown according to the colormap. The highlighted areas show the relative significance of these regions for the image classification. It can be seen from Fig. 14 that the image classification can be validated using these visualization techniques. The results also show that similar to better accuracies, better visualizations can be obtained with large models, such as VGG-19 compared to MobileNetV2.



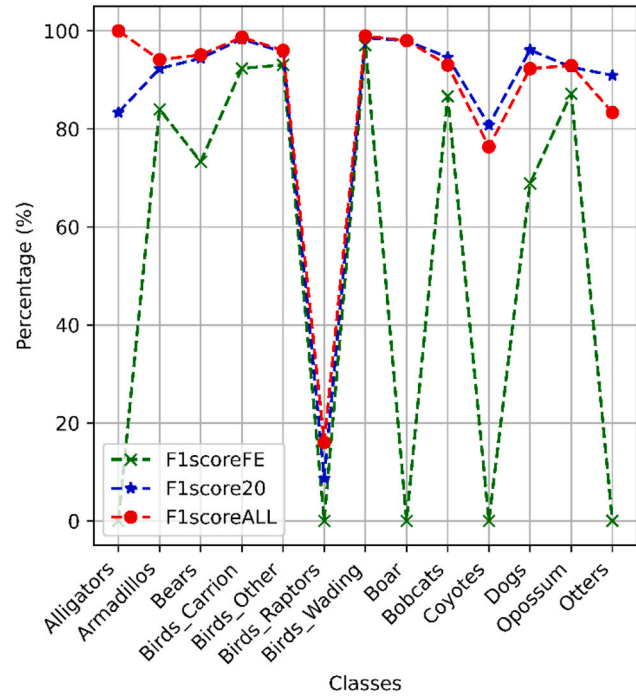
(a) Accuracy



(b) Precision



(c) Recall



(d) F1-Score

Fig. 10. Class-wise metrics for the Florida-Part1 dataset with MobileNetV2 model (a) Accuracy (b) Precision (c) Recall (d) F1-Score.

4.6. Power and latency measurements on RPi-Z

The prototype deployment used a weather-proof IP65 rated container shown in Fig. 15. The RPi-Z was connected to the solar powered battery and a PIR sensor fixed to the container. The idle current draw for RPi-Z was found to be around 70 mA, with a camera and PIR

sensor attached, measured using the Drok USB Tester. The current draw was further reduced to 50 mA by shutting down the HDMI corresponding to 0.25 W at 5 Volts.

The chargeable supply used for the prototype was rated at 900000 mAh which meant that the prototype camera trap could ideally be operated with 50 mA, for 18,000 h. However, the actual battery

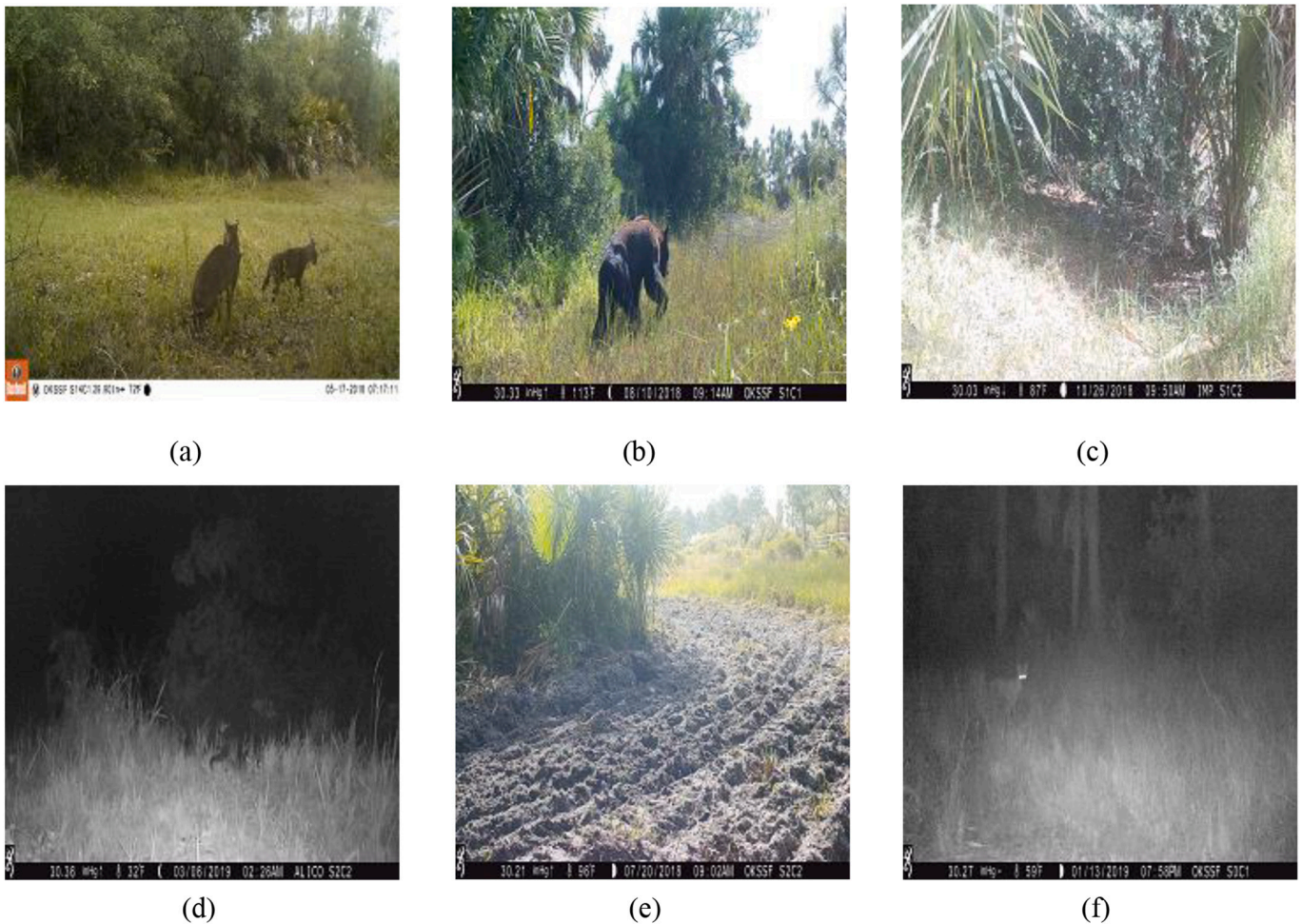


Fig. 11. Classification with the FineTuned-All model (a) Actual: Bobcats, Predicted: Bobcats (b) Actual: Bears, Predicted: Bears (c) Actual: Birds_Other, Predicted: Birds_Other (d) Actual: Otters, Predicted: Coyotes (e) Actual: Birds_Carrion, Predicted: Birds_Carrion (f) Actual: Coyotes, Predicted: Coyotes.

performance could be much lower due to losses and similarly the built-in solar panel performance is also governed by the sunlight available. The RPi-Z was run for 72 h as a test but the actual battery life would be different and can be determined under field conditions.

The processing time for running a prediction on an image from Florida-Part1 with MobileNetV2 was 6 s. However, the time for creating a Grad-CAM visualization with tf-explain (tf-explain, n.d) was 23.26 s. The visualization time was high due to the large number of computations performed for creating the visualization compared to only the model prediction.

5. Discussion

We have examined different transfer learning techniques for various DNN models suitable to run on a prototype camera trap for on-board animal classification, and determined that training more layers of a pre-trained model can improve the performance. The model performance was also shown to increase through the use of hyperparameter optimization techniques. We also investigated the use of on-board XAI techniques that can help in understanding the model decisions.

5.1. Transfer learning approaches

5.1.1. Performance of transfer learning approaches

Due to the high performance of the pre-trained models and their wide and open accessibility, it does not seem pragmatic to develop a customized DNN model any more, but rather to focus on performance

improvement with pre-trained model customization. A pre-trained model, trained on a large dataset provides an advantage of transfer learning to be used for classification on another dataset and is especially useful as no training will be required. The pre-trained model can also be fine-tuned by training some layers with another dataset, in either case resulting in a quicker training compared to designing and training the model from scratch. A DNN consists of millions of parameters and therefore requires large datasets to be optimally trained to avoid over-fitting, that is learning or memorizing the features from a smaller dataset, resulting in poor generalization. One alternative for transfer learning not considered in this study is to use a large camera trap dataset to fine-tune a pre-trained model on ImageNet. This trained model can then be used for transfer learning on a small camera trap dataset.

The transfer learning for the TensorFlow models used the three possibilities, feature extraction, fine tuning with some layers trained, and fine tuning with all layers trained. The results have shown an improvement in the classification performance with all layers trained. However, it is important to note that with feature extraction most of the pre-trained learning is preserved and used on newer tasks. The feature extraction was also seen as achieving high accuracy especially for small datasets with a reduced training time (Simoes et al., 2023; Willi et al., 2019). A performance improvement of MobileNetV2 model was reported with the model fine-tuning (Cunha et al., 2021). Similarly, depending on how many layers are trained in fine tuning, this can affect the model performance, with fine tuning of more layers resulting in learning of more features from the new dataset. In comparison to other studies (Carl et al., 2020; Duggan et al., 2021; Whytock et al., 2021) that

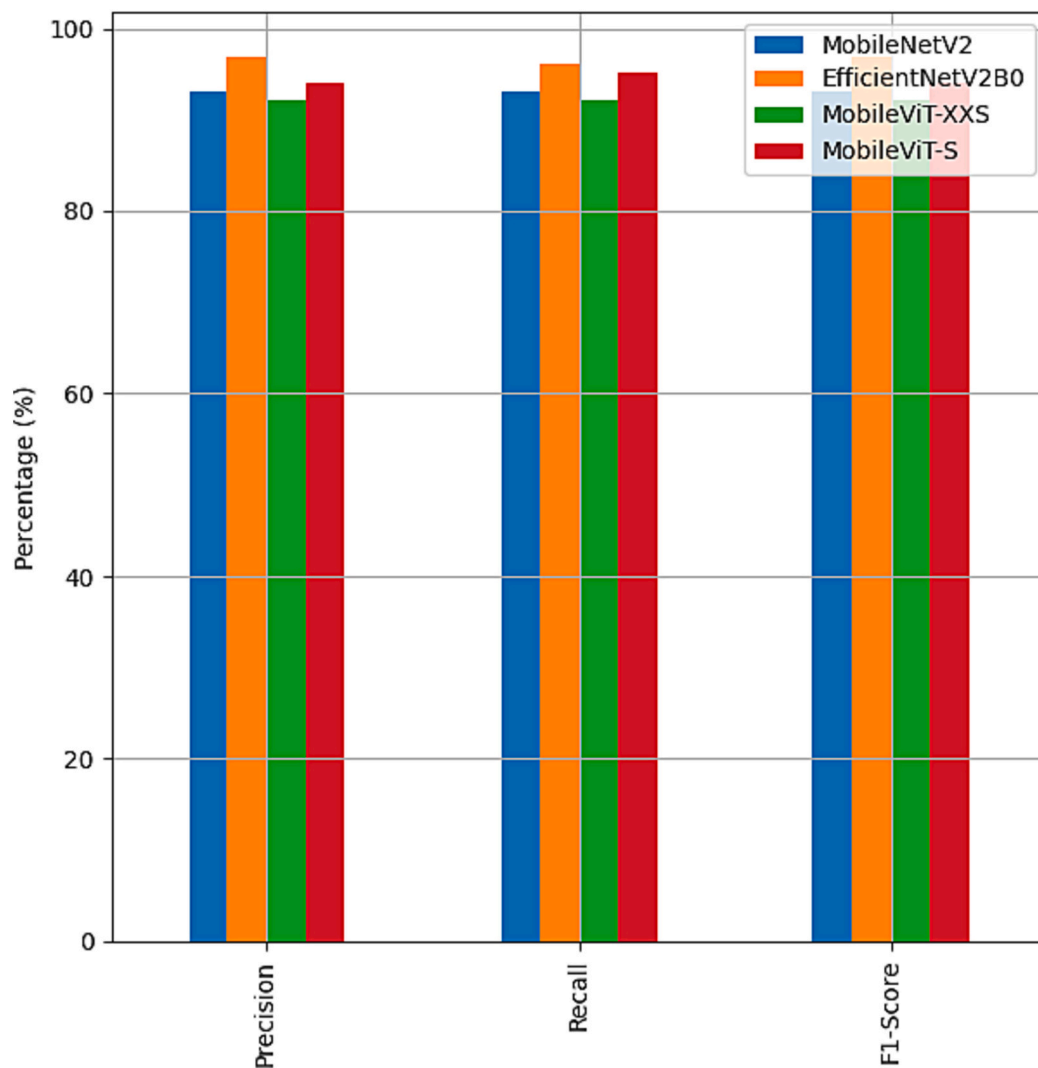


Fig. 12. Model Performance Comparison for the four selected models.

have used transfer learning primarily as feature extraction for classification, the results provided in this paper show an improvement in model's performance. Although the classification accuracy should not be compared across different datasets, however, compared to an accuracy of 92% for balanced dataset (Duggan et al., 2021), the results in section 4 show significant improvement with an unbalanced dataset used in this study with a simpler MobileNetV2 model. The use of DNN models with unbalanced datasets were also shown to have a comparatively lower classification accuracy of 35.4% (Villa et al., 2017).

In contrast to this study, feature extraction was seen as a better choice for animal classification, as the improvements with training all the layers of a pre-trained model were not found to be significant (Shahinfar et al., 2020). This could be due to the use of different and balanced datasets, and different models, and as only aggregate metrics were provided, therefore we cannot provide a comparison. However, as we have used four DNN models and used two datasets that were representative of the actual class distribution, we can conclude that in general the results of fine tuning would be better compared to feature extraction.

The real-world camera trap images are often challenging for classification tasks due to real-world field conditions (Liu et al., 2023). The performance of a DNN model depends to a very large extent on the training data quantity and quality that is used to train the model. Some automated systems face a major limitation of requiring high-quality

images, which is not always feasible. The dataset that we employed for conducting the classification performance was a challenging dataset, comprising many images that would have been difficult to be classified even by an expert. The research studies involving the camera trap images often use a subset of the entire dataset, create a balanced dataset, or manually extract the animal in the image for the model training (Norouzzadeh et al., 2018; Westworth et al., 2022). Training on a large number of wild mammal species with annotated bounding boxes can improve the classification accuracy (Carl et al., 2020). A study investigated the use of six DNN models for image classification by using six sample sizes, from 10 to 1000 images per class, and concluded that the accuracy, precision, and recall had a logarithmic relationship with the number of training images (Shahinfar et al., 2020; Sun et al., 2017).

In the model comparison, we have used two models based on the MobileViT, a vision transformer model. The vision transformers have been improving since their introduction for image classification, however, these have not been used so far in the ecological studies. We used the smallest of the vision transformer models suitable for the low power devices, however, there are other more powerful vision transformer models available which can be considered for image classification tasks.

The model choice could be governed by the required classification performance, model inference time, programming skills, and the availability of model training time and compute resources. Large models are suitable for off-site classification and visualization, whereas applications

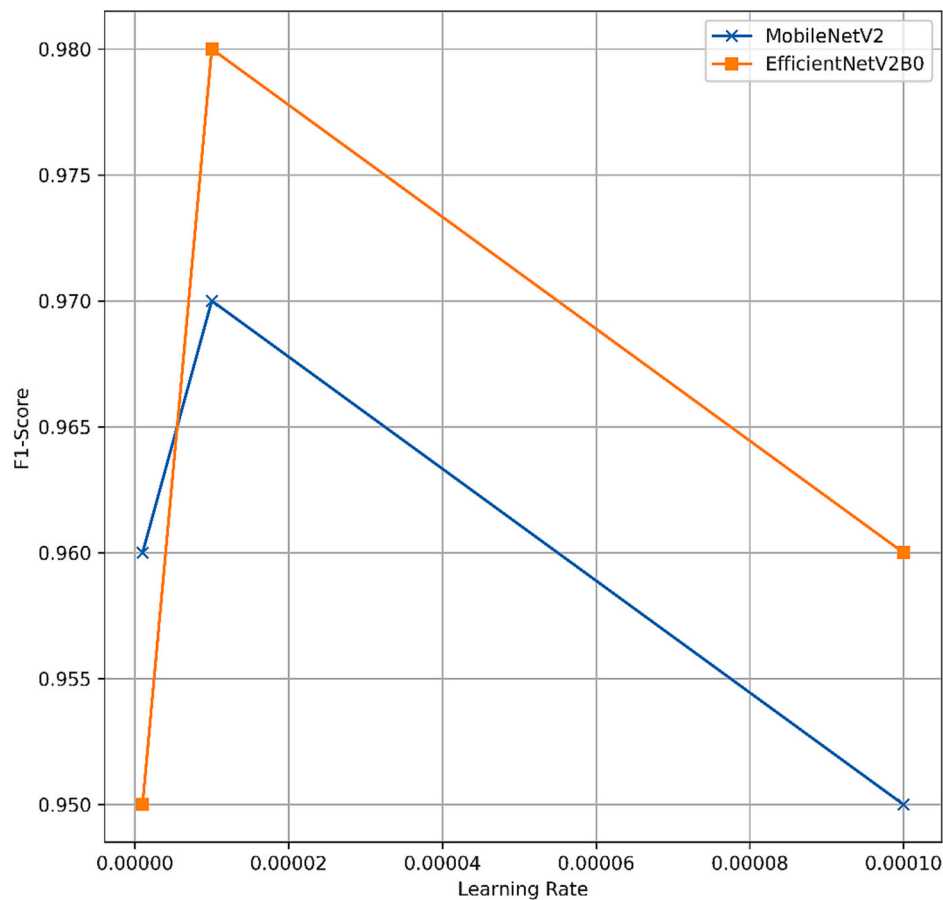


Fig. 13. Model Performance with three learning rates of 0.0001, 0.00001, and 0.000001 for the selected models with fine-tuning (all layers trained).

requiring classification in the field can use small sized models with the low power devices. Although transfer learning with fine tuning is used in some studies but this paper has investigated and provided a comparison that can be useful in making a model selection and its fine tuning. The model performance can also be further improved through the use of hyperparameter optimization.

5.1.2. Model generalization

A trained model on being presented with a new and unseen image should be able to predict the animal class, based on the image features it has learnt during the model training. The model generalization, that is, for a model to predict the new data from another site, is expected to perform similarly to the original test data, but this can only occur if the new site data is similar to that used during the model training. If the images are dissimilar, then the model will simply fail to recognize the animal not in the training dataset, or will perform badly. The large datasets such as ImageNet were created to benchmark the computer vision algorithms and comprise of various image classes, which are nested in each category. Development of a similar image dataset for ecology created by organizing the images into various animal classes can help train a universal model. There are frameworks already developed that can automate the process (Velez et al., 2023).

A dataset was created with a view to explore the generation of the trained model thereon to generalize to other locations (Beery et al., 2019). Similarly, the problem of generalization of the DNN models to out-of-sample camera trap images was investigated by using images from other camera trap sites, of various resolutions, and different camera trap models (Whytock et al., 2021). For a model to generalize to new data, the only requirement is for it to be trained with the images that it is expected to classify during model deployment. However, this would be

difficult and can work only if the new test images comprise of the same animal species under similar lighting conditions, and background etc. For long term studies, it could also be important to monitor the model's performance over a period of time and then re-train the model on the newly acquired camera trap images.

The problem of generalizability can also be solved by using incremental learning approaches that can help the model to retain the learnt classes (Zhu et al., 2022). The proposed approach was shown to overcome the problems associated with a naïve approach, of the bias and catastrophic forgetting. In catastrophic forgetting, the model's performance on the old classes deteriorates rapidly, which was alleviated using the proposed Centre Vector Retrieval (Zhu et al., 2022).

5.2. Edge device as camera trap

5.2.1. Classification and visualization

The object classification and visualization on the computing platform has been demonstrated using RPi-Z which has sufficient power to run DNN model predictions. Although we demonstrate this using a RPi-Z device, however, it is also possible to run the trained DNN models on even smaller and low-power devices, such as, microcontrollers. The advancements in image acquisition are not limited to ground based fixed deployments, such as camera traps, but increasingly drones (Petso et al., 2021), robotics, and satellites are also being used. The small models such as MobileNetV2 and TensorFlow Lite framework are designed for low power devices, such as mobile phones. The TensorFlow Lite model, in addition to running on RPi-Z, can run without changes on the mobile phone devices.

The ecologists are yet to fully adopt an automated workflow of image classification with DNN models (Whytock et al., 2021). The XAI can be

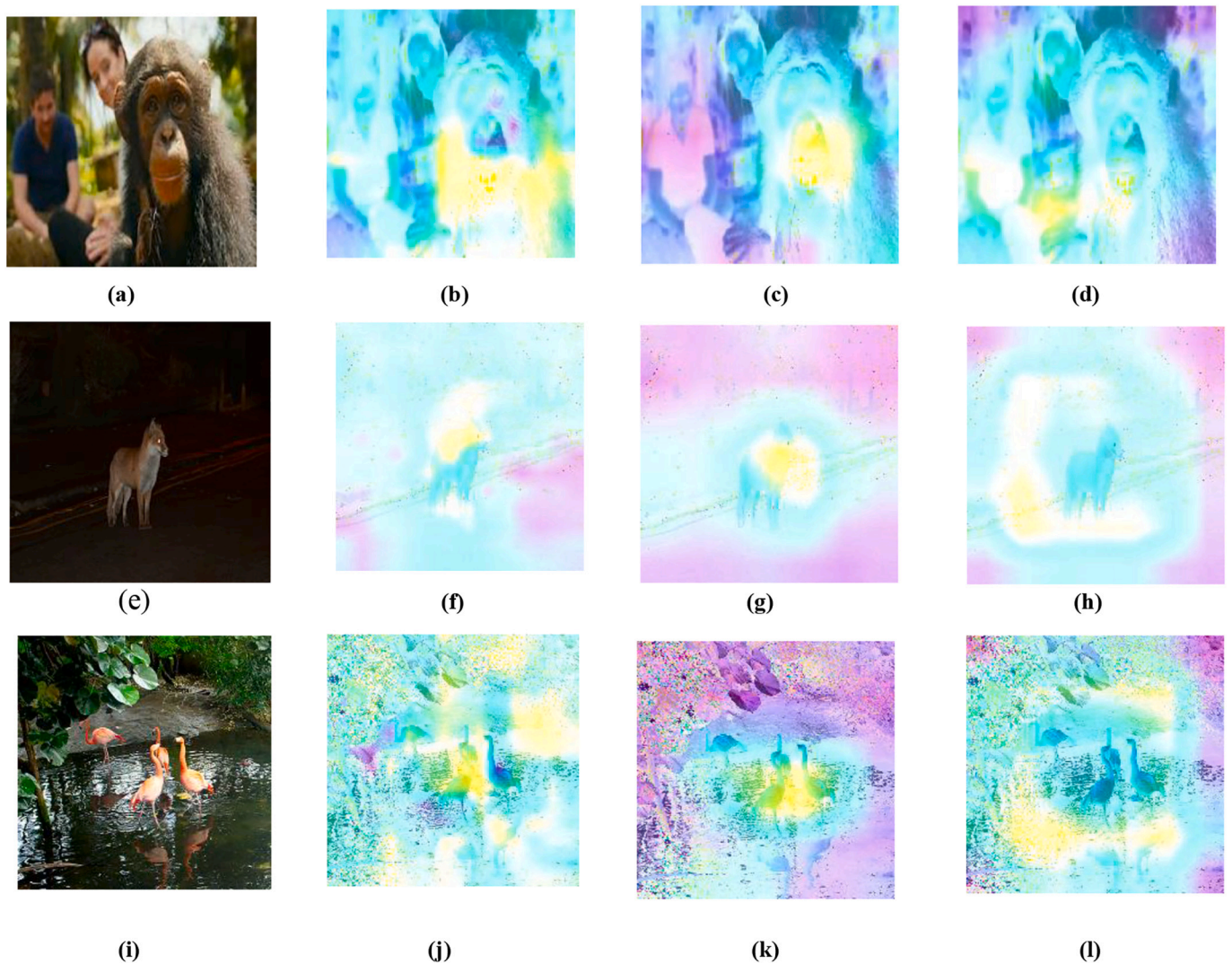


Fig. 14. Grad-CAM heatmap visualizations with VGG-19 on Google Colab and MobileNetV2 on RPi-Z (a) Original Chimpanzee image from ImageNet (b) Occlusion Sensitivity with VGG-19 (c) Grad-CAM with VGG-19 (d) Grad-CAM with MobileNetV2 (e) Original Red Fox image from ImageNet (f) Occlusion Sensitivity with VGG-19 (g) Grad-CAM with VGG-19 (h) Grad-CAM with MobileNetV2 (i) Original Flamingo image from ImageNet (j) Occlusion Sensitivity with VGG-19 (k) Grad-CAM with VGG-19 (l) Grad-CAM with MobileNetV2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

used to interpret the wildlife images and can be used to evaluate the performance of the DNN models. These techniques could be used together with object detection (Carl et al., 2020), or segmentation techniques to accurately identify an animal in the image and would be helpful especially for challenging datasets (Giraldo-Zuluaga et al., 2017). This could also be used to investigate where a DNN is focusing attention for its classification decision. For example, to identify the same species across different locations and a different background (Beery et al., 2019).

A customized camera trap opens many opportunities for utilizing the software and hardware components providing complete control on what is possible compared to the commercial camera traps. The hardware and software of the open access camera trap, WiseEye was configurable, providing on-board image processing and cellular communications, but required a large battery and solar panel (Nazir et al., 2017b). Such devices can also use satellite networks for a truly global connectivity. The real-time classification of images and off-site communication can help facilitate timely and informed decisions for preserving the environment and ecosystem.

The use of open access visualization techniques for image

classification are readily available and can be easily integrated. Some studies have reported the use of explainable techniques for understanding the model (Ryo et al., 2021) but this study provides the running of the XAI techniques on a prototype camera trap for animal classification.

5.2.2. Energy savings on the camera trap edge device

The running of ResNet-18 (224×224) with TensorFlow Lite on RPi 3 was shown to take 4.03 s (Mathur and Khattar, 2019). TensorFlow Lite with optimizations on latency and size of the trained model can provide benefits, however, for RPi based implementations, it is useful to utilize smaller models, such as MobileNet. This will have some loss in accuracy compared to the larger models, such as ResNet-50 (Mathur and Khattar, 2019), but the benefit is a smaller model size and faster inferences on the edge device. Similarly, the model weights can be quantized for the models to have a significant reduction in memory usage compared to the floating-point models (Cunha et al., 2021), however, this would also be accompanied by some loss in the model's accuracy.

A camera trap with energy harvesting could be always on for continuous video monitoring to determine real-time events, with an



Fig. 15. A prototype camera trap device based on Raspberry Pi Zero (RPI-Z).

additional advantage that PIR triggers, susceptible to an event capture loss could be avoided (Nazir et al., 2017b; Riechmann et al., 2022). With continuous video monitoring there are great advances to be exploited in this area. Most cameras use Motion JPEG (MJPEG) or H.264 video compression standards. For example, H.264 video was processed to detect an event of interest (Riechmann et al., 2022). However, with the recent video compression advancements, efficient algorithms, such as High Efficiency Video Coding (HEVC) standard with much higher compression efficiency, and thus a lower communication bandwidth requirement, will become commercially available. Such advancements would help make it possible to transmit live video from the field devices equipped with a suitable power source.

Personal computers and microcontrollers have a sleep mode using which the power consumption can be reduced to a minimum, with a quick transition to the operational mode as required, without going through the start-up sequence. RPI-Z does not have a sleep mode but there are some components, such as, HDMI, Wi-Fi, Bluetooth on the RPI-Z board that may not be required for a field deployment, and can be powered down economising on the total power consumed in running the device (Droissart et al., 2021). The use of RPI-Z is well-suited for the edge AI applications, due to its low power requirements, low cost, and small size. However, this can be further facilitated for field use, with a sleep mode similar to laptops or microcontrollers. In future, it is expected that more powerful and miniaturised commercial models will emerge.

The advantage of on-board processing compared to off-loading the images off-site is that compared to communications, storage and processing require much less power (Nazir et al., 2017a; Glover-Kapfer et al., 2019). Any application that requires to conserve battery power can reduce the communications cost by transmitting less information. The images were communicated from the camera trap through the Simple Mail Transfer protocol (SMTP) server (Chalmers et al., 2023). However, depending on the activity in the camera trap's PIR field of view can result in many images communicated, especially the false positives. The communication cost can be kept low by sending the extracted information of interest rather than the image data. The confidence levels and the class label were transmitted over Long-range Wide Area Network (LoRAWAN) to a cloud platform to conserve the battery power (Zualkernan et al., 2022). The WiseEye camera trap used confirmatory sensing by using a radar and on-board image differencing technique to improve the PIR based animal detection to help segregate the empty images, and employed cellular communication for communicating the image data (Nazir et al., 2017b).

The satellite terminal and the communications from the remote site were controlled by a RPi to autonomously utilize and conserve power as required (Nazir et al., 2017a). For applications requiring image data, communication of a thumbnail image from the Joint Photographic Experts Group (JPEG) image or subsets of the image data can be considered for communications from the remote site (Nazir et al., 2020b). For the processing on the embedded platforms it is convenient to use smaller and optimized models on relatively low form factor devices such as RPI-Z to economize power.

The data communications can take place from virtually anywhere in the world using the satellite communication module integrated with a Raspberry Pi or similar device, however, communications costs are substantially more compared to cellular networks. A satellite module was used to transmit image labels from a RPi device linked with a commercial camera trap for image capture (Whytock et al., 2023). The setup used a solar panel and the RPi could be powered off once not communicating (Whytock et al., 2023). Sending a thumbnail image would cost \$2 USD per image on the Iridium network with a monthly contract (Whytock et al., 2023). Hence it is important to reduce the data size by sending the image labels, or thumbnail images only based on the application needs.

5.3. Study uncertainties and limitations

The model evaluation was performed on two datasets which are in comparison much smaller to the ImageNet dataset used for the pre-trained models. Hence the proposed techniques of fine tuning have not been applied to the large datasets. However, the general rules provided in section 3.3 for selecting a particular transfer learning (just replacing the classification head vs. fine-tuning by training few or all layers) can guide the implementation choice. The selection of an optimum model however would also be governed by experimentation, to resolve uncertainty in the process.

Hyperparameter optimization was illustrated with only a single dataset and one hyperparameter. Such limited explorations could be useful unless the aim is to optimize a single model by considering all the important model hyperparameters (Kuncheva et al., 2023). The hyperparameter tuning was shown to improve the model performance (Fig. 13) for the selected models. Only a single hyperparameter, that is, the learning rate was optimized with hyperband and Bayesian Optimization techniques. In actual practice, more hyperparameters would need to be considered to design an optimized model. Another factor to consider is that different models have different number and types of hyperparameters, and trying to fully optimize each will require considerable compute resources and time as the hyperparameter search space would be large.

The performance of a model can be improved using hyperparameter optimizations but the performance of a model also depends to a very

large extent on the training data. Thus, it is not certain that the model with the best performance on one dataset, will also perform best on another dataset. This is especially true for the pre-trained models where the reported performance metrics are linked to a particular dataset, for example, MobileNetV3 with a reported higher accuracy than MobileNetV2, did not perform better (Yang et al., 2022). The performance of MobileNetV2, EfficientNetB0, and EfficientNetB3 were compared on the Caltech camera traps dataset and Snapshot Serengeti, and the EfficientNetB3 was found to perform better for classification (Cunha et al., 2021). The proposed model's performance was improved by making changes to the MobileNetV3 architecture, and was shown to perform better than the MobileNetV2 and MobileNetV3 architecture, for bird song classification (Yang et al., 2022). Similarly, the number of MobileNetV2 model filters were reduced to reduce the latency (Cunha et al., 2021).

6. Conclusion and future work

In this paper, we have shown that through transfer learning with fine tuning, the performance of state-of-the-art deep learning models can improve animal classification results in the field on-board low power customized camera trap on edge devices. The Deep Neural Network model selection has the trade-offs concerning the model size, model performance, model training and inference time, computations complexity, and ease of use. Therefore, the studies can benefit by considering these trade-offs to select a model depending on the application performance requirements. The model comparison showed that EfficientNetB0 had the best performance amongst the selected small sized models. The need for a human-in-the-loop would always remain to validate the model performance and predictions.

The use of hyperparameter optimization demonstrated the model improvement and needs to be carefully considered. The improvements with the model fine tuning and hyperparameter optimization are applicable to any DNN model. The classification and visualization process flow on a Raspberry Pi in a camera trap can also be extended to other use cases and scenarios, such as, image acquisition using a drone,

extending the possibility for a wide coverage. The improvements in Explainable Artificial Intelligence techniques can make it easy to understand and interpret the deep learning models' predictions, and also to evaluate and improve the models. The large-scale inferences from a study could see an interplay between the low power edge devices and virtually unlimited storage and compute resources with cloud technologies. By incorporating the processing on the camera trap, it is possible to obtain timely information from large scale ecological studies, even at the global scale. With further improvements in the battery technologies, deep learning model optimizations, and better hardware, it would soon become easier and common for ecological studies to utilize customized camera trap devices with on-board classification, visualization, and communications.

In future work, we will investigate model optimization techniques for deploying the deep learning models to even smaller microcontroller devices. Such low power devices can further economize the batteries and run autonomously to provide near real-time ecological insights from the field. We would also research other model fine-tuning techniques to improve the model performance by investigating model pre-training on large datasets from ecological studies.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Appendix

Table A1
Class wise statistics for the Florida-Part1 dataset.

Technique	Metric	Image Class												
		Alligators	Armadillos	Bears	Birds_Carrion	Birds_Other	Birds_Raptors	Birds_Wading	Boar	Bobcats	Coyotes	Dogs	Opossum	Otters
		7	26	103	270	490	15	765	25	493	30	40	281	6
Feature extension	TP	0	21	81	260	468	0	731	0	461	0	21	239	0
	FP	0	3	37	33	48	0	10	0	110	0	0	28	0
	TN	2544	2522	2411	2248	2013	2536	1776	2526	1948	2521	2511	2242	2545
	FN	7	5	22	10	22	15	34	25	32	3	19	42	6
	Accuracy (%)	99.73	99.69	97.69	98.31	97.26	99.41	98.28	99.02	94.43	98.82	99.26	97.26	99.76
	Precision (%)	0.00	87.50	68.64	88.74	90.70	0.00	98.65	0.00	80.74	0.00	100.00	89.51	0.00
	Recall (%)	0.00	80.77	78.64	96.30	95.51	0.00	95.56	0.00	93.51	0.00	52.50	85.05	0.00
	F1-score (%)	0.00	84.00	73.30	92.36	93.04	0.00	97.08	0.00	86.65	0.00	68.85	87.23	0.00
	TP	5	24	93	266	477	1	751	24	466	21	37	270	5
	FP	0	2	1	5	30	7	7	0	26	1	0	32	0
FineTuned-20	TN	2544	2523	2447	2276	2031	2529	1779	2526	2032	2520	2511	2238	2545
	FN	2	2	10	4	13	14	14	1	27	9	3	11	1
	Accuracy (%)	99.92	99.84	99.57	99.65	98.31	99.18	99.18	99.96	97.92	99.61	99.88	98.31	99.96
	Precision (%)	100.00	92.31	98.94	98.15	94.08	12.50	99.08	100.00	94.72	95.45	100.00	89.40	100.00
	Recall (%)	71.43	92.31	90.29	98.52	97.35	6.67	98.17	96.00	94.52	70.00	92.50	96.09	83.33
	F1-score (%)	83.33	92.31	94.42	98.34	95.69	8.70	98.62	97.96	94.62	80.77	96.10	92.62	90.91
	TP	7	24	97	263	481	2	753	25	447	21	36	276	5
	FP	0	3	37	33	48	0	10	0	110	0	0	28	0
	TN	2544	2522	2411	2248	2013	2536	1776	2526	1948	2521	2511	2242	2545
	FN	7	5	22	10	22	15	34	25	32	3	19	42	6
	Accuracy (%)	99.73	99.69	97.69	98.31	97.26	99.41	98.28	99.02	94.43	98.82	99.26	97.26	99.76
FineTuned-All	TP	7	24	97	263	481	2	753	25	447	21	36	276	5
	FP	0	3	37	33	48	0	10	0	110	0	0	28	0

(continued on next page)

Table A1 (continued)

Technique	Metric	Image Class												
		Alligators	Armadillos	Bears	Birds_Carrion	Birds_Other	Birds_Raptors	Birds_Wading	Boar	Bobcats	Coyotes	Dogs	Opossum	Otters
		7	26	103	270	490	15	765	25	493	30	40	281	6
	FP	0	1	4	0	31	8	4	1	21	4	2	37	1
	TN	2544	2524	2444	2281	2030	2528	1782	2525	2037	2517	2509	2233	2544
	FN	0	2	6	7	9	13	12	0	46	9	4	5	1
	Accuracy (%)	100.00	99.88	99.61	99.73	98.43	99.18	99.37	99.96	97.37	99.49	99.76	98.35	99.92
	Precision (%)	100.00	96.00	96.04	100.00	93.95	20.00	99.47	96.15	95.51	84.00	94.74	88.18	83.33
	Recall (%)	100.00	92.31	94.17	97.41	98.16	13.33	98.43	100.00	90.67	70.00	90.00	98.22	83.33
	F1-score (%)	100.00	94.12	95.10	98.69	96.01	16.00	98.95	98.04	93.03	76.36	92.31	92.93	83.33

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